Analisi Strutturale e Modellazione di Ponti ad Arco in Muratura

Studio di un caso pratico applicato alla viabilità bolognese

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Preparation of this work was facilitated by the use of the *IPLeiria-Thesis* template.

Acknowledgements

Writing Guidance

In the *Acknowledgment* section, express your gratitude to those who helped and supported your work. Start by thanking your advisors, mentors, or supervisors who provided guidance and expertise. Mention any colleagues, classmates, or team members who contributed to discussions or offered assistance. You can also acknowledge specific organisations, institutions, or funding sources that supported your research or work. Lastly, include any personal acknowledgments for family or friends who offered encouragement and moral support during the project. Keep this section sincere, concise, and professional.

Resumo

Writing Guidance

Na secção *Resumo*, apresente um resumo conciso do seu projeto, destacando os pontos principais. Comece com uma breve declaração do problema ou objetivo, seguido de uma descrição da sua abordagem ou metodologia. Resuma os principais resultados ou conclusões, salientando a sua importância ou implicações. Conclua com uma ou duas frases sobre a contribuição global ou o impacto do seu trabalho. O resumo deve ser claro e conciso, idealmente com 150-250 palavras, para que os leitores compreendam rapidamente o seu trabalho e a sua importância.

Palavras-Chave: Palavra-Chave A, Palavra-Chave B, Palavra-Chave C.

Abstract

Writing Guidance

In the *Abstract* section, provide a concise summary of your project, highlighting the key points. Begin with a brief statement of the problem or objective, followed by a description of your approach or methodology. Summarise the main results or findings, emphasising their significance or implications. Conclude with a sentence or two on the overall contribution or impact of your work. Keep the abstract clear and focused, ideally within 150-250 words, to give readers a quick understanding of your research and its importance.

Keywords: Keyword A, Keyword B, Keyword C.

AI Acknowledgement

Writing Guidance

It is good academic practice to briefly state how, why, and when AI was used. Explaining both the actions taken and the reasons behind them encourages critical reflection, deepens understanding of AI's role in learning, and supports the evaluation of its impact on your work's quality and integrity. This page can be removed by setting the option `aiack` to `false` in the document class options.

Example of Use

I acknowledge the use of ChatGPT (`https://chatgpt.com`) to refine the academic tone and improve the linguistic accuracy of this work, including aspects of grammar, punctuation, and vocabulary.

Description of Use
Prompt 1: In a paragraph, simply describe what LaTeX is.
Output 1: LaTeX is a high-quality typesetting system commonly used for producing scientific and mathematical documents due to its powerful handling of formulas and bibliographies. Unlike word processors, it uses plain text files with markup tags to define the structure and formatting of a document, allowing precise control over layout and appearance. It is especially popular in academia and research for creating theses, journal articles, and presentations.

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1

Introduction to the Template Motivation and First Steps

Author: José Areia
Current Version: 2.2.13
License: LaTeX Project Public License v1.3c
Official Repository: GitHub Repository

Welcome to the *IPLeiria Thesis* template! Thank you for choosing it for your dissertation, report, or project. This template reflects many hours of development, and I hope you enjoy using it as much as I did creating it. This chapter introduces its purpose and helps you get started. See **??** for a detailed guide, and **??** for a brief LaTeX tutorial to maximise its use.

1.1 Motivation

I've been using LaTeX since 2020 for a wide range of purposes. Over time, I've reviewed over a hundred templates, and there's always something missing. *Always*. Powerful templates—*i.e.*, highly customisable with many options—are often poorly organised. Well-organised ones usually lack flexibility. Some even compile with errors and warnings. Most importantly, many aren't user-friendly. So, I created my own template for theses and reports, tailored to the Polytechnic University of Leiria. My goals were: i) clear and structured file organisation, ii) a clean, professional, and attractive design, iii) high customisability, and iv) ease of use, especially for beginners.

1.2 Getting Started

To start using this template, you first need to know how to use LaTeX. For this, please refer to **??**. Once you are familiar with LaTeX, you will need to either install it locally or use an online LaTeX editor.

If you prefer an online editor, I highly recommend Overleaf. While Overleaf offers a paid subscription for extended compilation time, this template is specifically designed to compile within the limits of the free subscription plan. To use the template in Overleaf, just refer to official template page and click *Use as Template*. But if you prefer to use a different version of it (which I do not recommend), you can do the following:

1. *Download the desired version from the GitHub repository as a Zip file.*
2. *Login to your Overleaf account.*
3. *In your Project area, click in the menu and then: New Project → Upload Project.*
4. *Upload the Zip file.*
5. *Let Overleaf compile the document.*

If you choose to use a local editor, you must first install LaTeX on your machine. For this, there are several options, but I personally recommend either TeX-Live or MikTeX. After installing LaTeX, you will need to select an editor for writing and editing your documents. To help with this decision, I suggest checking out this helpful post, which provides a comprehensive overview of various editors you can use. Once LaTeX and your editor are set up, simply clone or download the latest version of the template from GitHub and start using it!

> **Tip**
>
> In the official GitHub repository, you'll find a Makefile and a Latexmk configuration file, either of which can be used to compile the project. The Makefile supports both `rubber` and `latexmk` as compilers, so you can choose the one that best suits your preferences.

1.3 Getting Help

As a newcomer, you may encounter situations where you want to do something in LaTeX or with this template but aren't sure how. When questions arise, you have several options. First, you can read the wiki available in the GitHub repository for this template. Another great option is the TeX Stack Exchange, an active community that can help with nearly any issue. Of course, Google is always a reliable resource. If all else fails, feel free to contact me directly with any questions about the template. You can reach me at *jose.apareia@gmail.com*.

1.3.1 Issues, Feature Request and Suggestions

If, by any chance, you encounter a bug, have a suggestion, or would like to request a feature, you can submit them via the *Issues* tab in the GitHub repository. Please be as de-

scriptive as possible when reporting issues, and make sure to provide the appropriate labels to help me triage them effectively.

For feature requests or suggestions, you can either follow the steps mentioned above or, if you prefer, you can implement the feature yourself and submit a pull request. Both pull requests and pushes trigger a GitHub Action that will automatically compile the document. If the compilation fails, the pull request will be automatically rejected. Please keep this in mind and take care when submitting changes.

1.3.2 In-Built Comments, Guidance Texts and Warnings

Within this document template, you may encounter informational text displaying the message "Writing Guidance." These sections are provided solely as guidance to help users understand what content should be included in specific sections. They are not related to the LaTeX code itself within this template.

While navigating through the template, especially the configuration files, you will notice that everything is thoroughly commented. LaTeX can sometimes be difficult to understand without proper documentation for the packages we are working with. With that in mind, I have made an effort to comment on all the changes I've made. Occasionally, I may forget or deem it unnecessary to comment on simpler changes, but more advanced modifications are always accompanied by detailed comments.

Finally, regarding warnings: please take them into consideration when compiling the document. As you may have noticed, the template is designed to be free of warnings, so please strive to maintain this clean compilation.

1.4 Important Notices

Although this template is specifically tailored for students from all six schools of the Polytechnic University of Leiria, you are welcome to use it if you are from a different institution. It is highly customisable and can be easily modified to suit the needs of other schools. If your school is interested in adopting this as the official template, please contact me first.

If you decide to use this template, please consider acknowledging it in your work. To do so, simply cite it using `\citep{IPLeiriaThesis}`. You can also show your appreciation for this template by giving a star on the GitHub repository. This helps increase its visibility, and you will receive notifications about the latest updates and releases. Either way, acknowledging this work, which involved significant effort and countless hours, would be greatly appreciated.

Appendices

A
Showcasing the First Appendix

Writing Guidance

Appendices contain supplementary material **created by the author** that enhances the reader's understanding of the dissertation while not being essential for following the primary narrative. These sections often include detailed tables, figures, complex calculations, or materials like survey questions and interview transcripts produced in the course of the research. The appendices allow readers to explore the research in greater detail, offering a deeper insight into methods and findings without interrupting the main body of work.

B

Showcasing the Second Appendix

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Annexes

L

Showcasing the First Annex

Writing Guidance

Annexes are supplementary sections in a dissertation that provide additional information or external documents not essential to the main arguments but that support or complement the research. Unlike appendices, **annexes generally contain material that was not developed by the author**, such as reports, legal documents, or published datasets from external sources. This information is placed separately to keep the main content concise, allowing readers access to relevant external references without disrupting the dissertation's flow.